

Reconstructing Partial Fingerprints Using PatchGAN Discriminator (NEXUS FORENSIC LAB)

*Reconstructing damaged fingerprint images using
autoencoders to enhance biometric identification accuracy.*

Team Members:

VISHESH

SHRUTHI

SHREYA

VISHNU

SREEJA

Introduction

- Fingerprints are crucial for secure authentication in areas like **law enforcement, banking, and mobile security**. However, real-world images often suffer from noise and distortions.
- This project uses U-net and PatchGAN to **enhance fingerprint quality by denoising, compressing, and extracting features**. The model learns to reconstruct cleaner images.
- We have built and trained our model from the ground up.

Introduction



Noisy



Reconstructed

Problem Statement

- Fingerprint images captured in real-world conditions are often **noisy, incomplete, or distorted** due to sensor limitations and environmental factors.
- Poor-quality fingerprints decrease biometric system accuracy, leading to more **false acceptances** and rejections.
- High-resolution fingerprint images require **large storage** and slow data transmission.
- Effective automated techniques are needed to enhance image quality, compress data, and extract robust features for fast, secure matching.

Business Use Case

Biometric Authentication Systems

- Enhances
- Improve identity verification
- Boost efficiency and robustness.
- Support real-time, reliable authentication.

Law Enforcement & Forensics

- Support forensic investigations efficiently.
- Improve accuracy
- Enable faster matching
- Denoise

Citizen ID & e-Governance

- Enable fast and secure verification.
- Enhance efficiency in e-governance services.
- Support programs like Aadhaar and passports.
- Optimize storage of fingerprint data.

Data Overview

Source:

Kaggle Dataset

FVC2002 public fingerprint dataset with grayscale images from multiple subjects under different capture conditions.

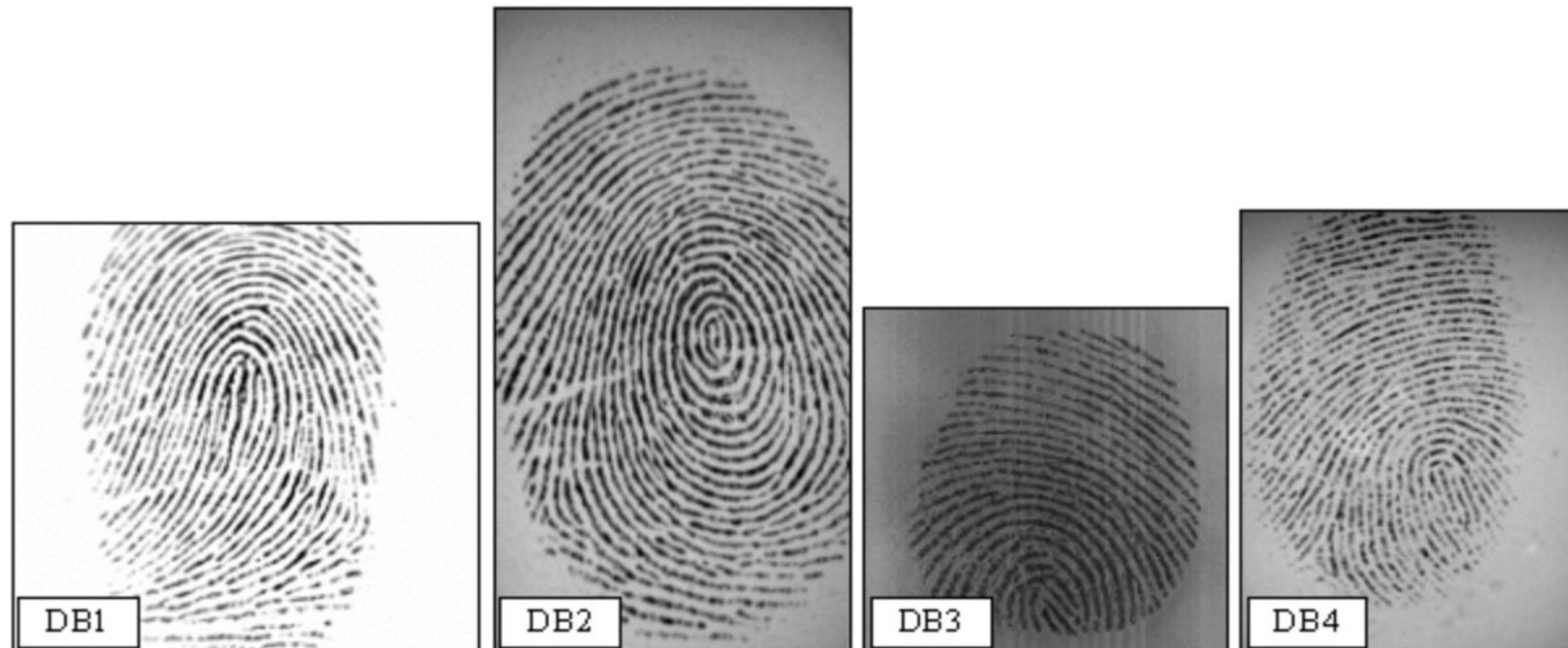
Dataset Characteristics:

This dataset consists of 4 different sensor fingerprints:

- Low-cost Optical Sensor
- Low-cost Capacitive Sensor
- Optical Sensor
- Synthetic Generator

Each sensor has varying image sizes. The dataset has 3200 images, 800 images per sensor.

Exploratory Data Analysis



Exploratory Data Analysis

1. Dataset Composition

- Multiple databases (DB1–DB4) with consistent fingerprint image counts.

2. Image Properties

- Grayscale `.tif` images with uniform dimensions and intensity distribution.

3. Sample Visualization

- Visual inspection of full and partial fingerprints for quality and variation.

4. File Naming & Structure

- Consistent naming scheme indicating finger ID and impression number; useful for organizing and grouping data.

Data Preprocessing

Load Fingerprint Images

Glob library is used to target all .tif files and **load the dataset** across the nested folders.

Processing images

All the images are read in **grayscale mode** and resized to **128 by 128 pixels**. Then, they are **normalised to range [0,1]** for convenient training. Normalised images are converted into **4d tensors (N, 128, 128, 1)** to match expected input of Keras models.

Noise Injection

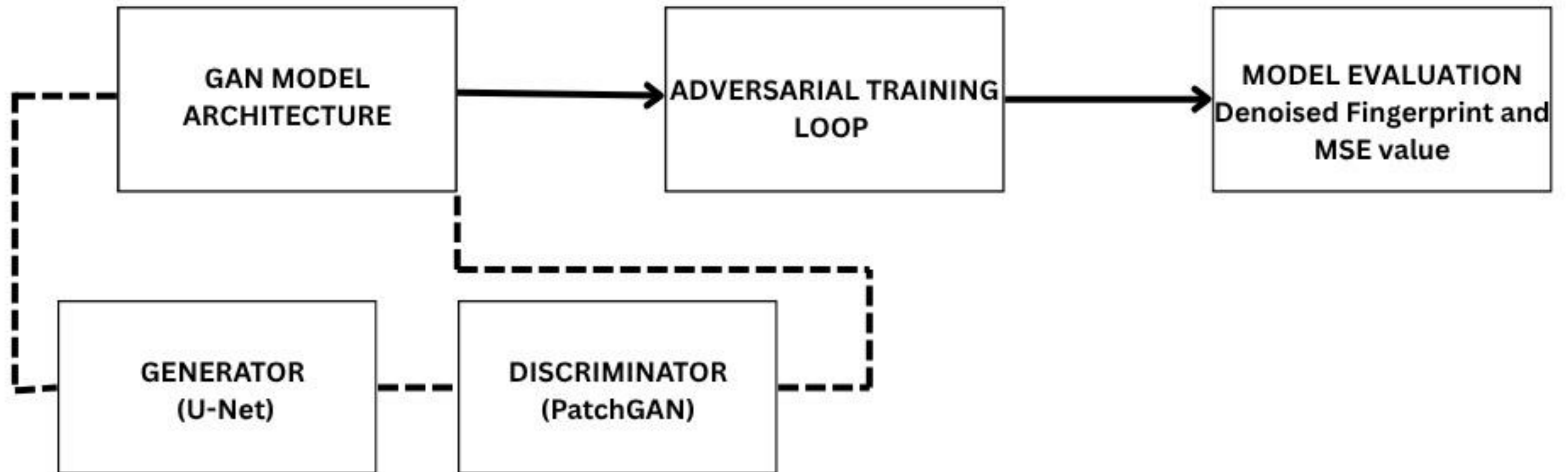
Gaussian noise blur is added in three different training stages to simulate real world noise in the input.

Final Dataset Building

Noisy images are again **clipped** to ensure the new images remain in the range of [0,1]. Then, the noisy images (real world input) and the original images (ground truth) are **matched** to create a **supervised dataset**.

Methodology

PatchGAN Discriminator



Methodology

1. GAN Model Architecture

- **Generator:** Uses a **U-Net** architecture with encoder-decoder structure and skip connections to reconstruct clean fingerprints from noisy inputs while preserving fine ridge details.
- **Discriminator:** Uses **PatchGAN**, which breaks the image into small patches and classifies as real or fake, focusing on local textures and ridge patterns crucial for fingerprint realism.

2. Adversarial Training Loop

- **Generator Objective:** Minimize **L1 reconstruction loss** and simultaneously **fool the discriminator**.
- **Discriminator Objective:** Differentiate real clean fingerprints from generated outputs.
- **Simultaneous Updates:** Both networks are trained together in an adversarial setup to improve reconstruction quality.

3. Model Evaluation

- **Qualitative:** Visually inspect denoised fingerprints to check restoration of ridge patterns.
- **Quantitative:** Evaluate using **Mean Squared Error (MSE)** and plot **generator loss across epochs** to monitor convergence and accuracy.

Results and Analysis

1/1 1s 757ms/step

Clean



Clean



Clean



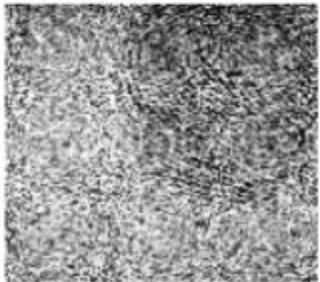
Clean



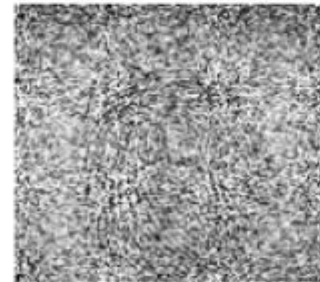
Clean



Noisy



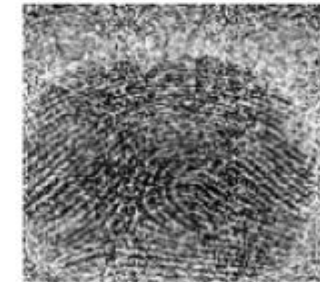
Noisy



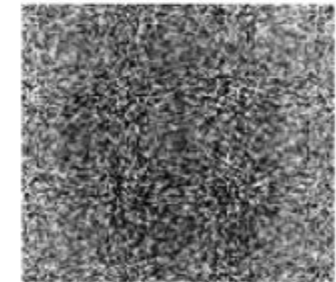
Noisy



Noisy



Noisy



Restored



Restored



Restored



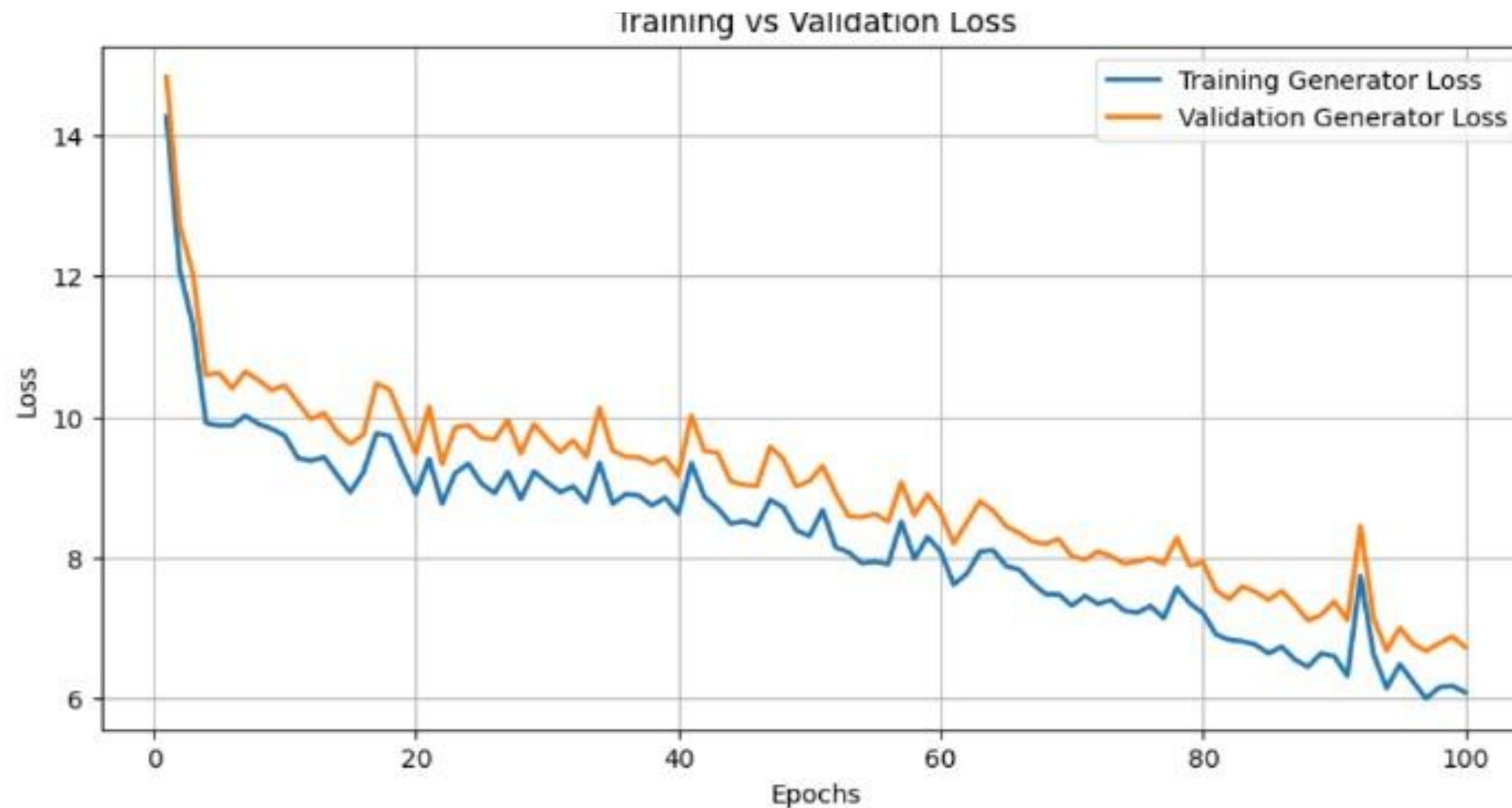
Restored



Restored



Result and Analysis



Mean Squared Error on 5 test samples: 0.005

Mean Squared Error on full dataset: 0.006

Mean Squared Error between clean and noisy images: 0.064

Live Demo

NEXUS Forensic Lab

• System Online • AI Model Ready

Fingerprint Enhancement Division

UPLOAD FINGERPRINT



Drag & Drop your .tif fingerprint image here
or click to browse

Live Demo

PROCESSING RESULTS

Original



Enhanced



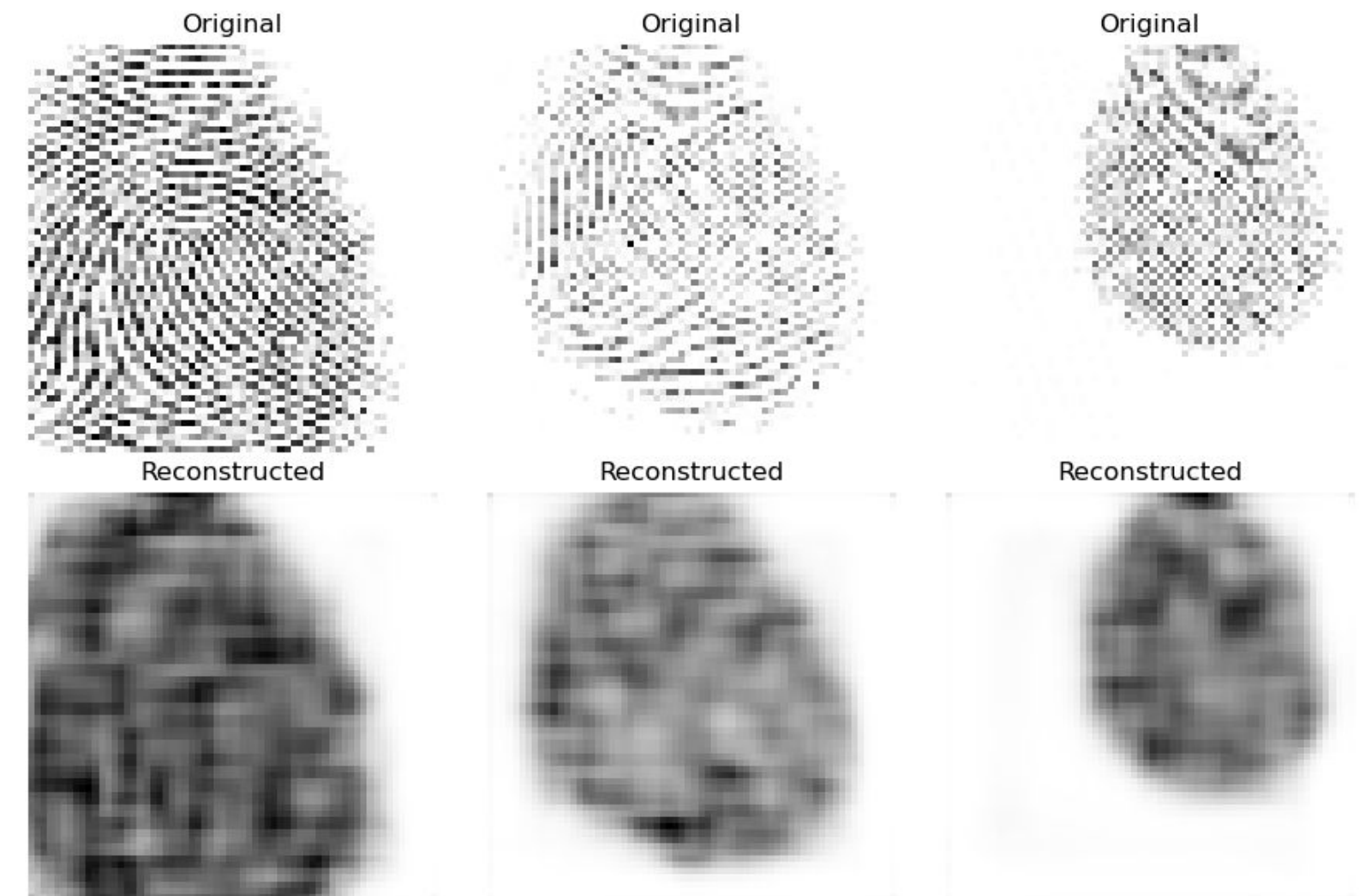
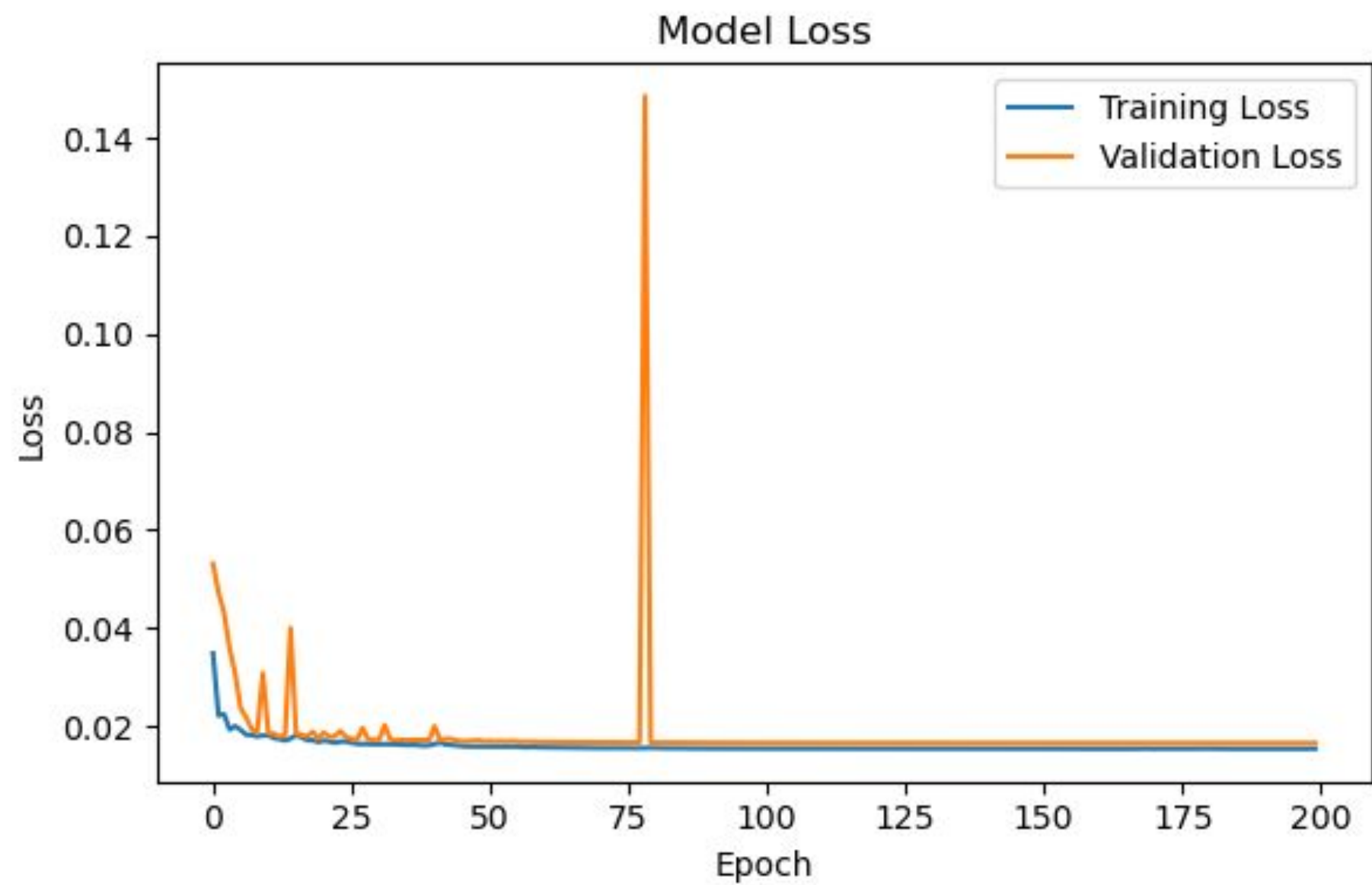
Challenges

- ❑ **Data Quality & Variability:** The dataset suffers from **low-quality, misaligned images** with variations in scale, rotation, and background noise, impacting model performance.
- ❑ **Security Risks:** Synthetic fingerprint reconstruction poses **potential misuse risks**, such as spoofing or bypassing biometric authentication systems.

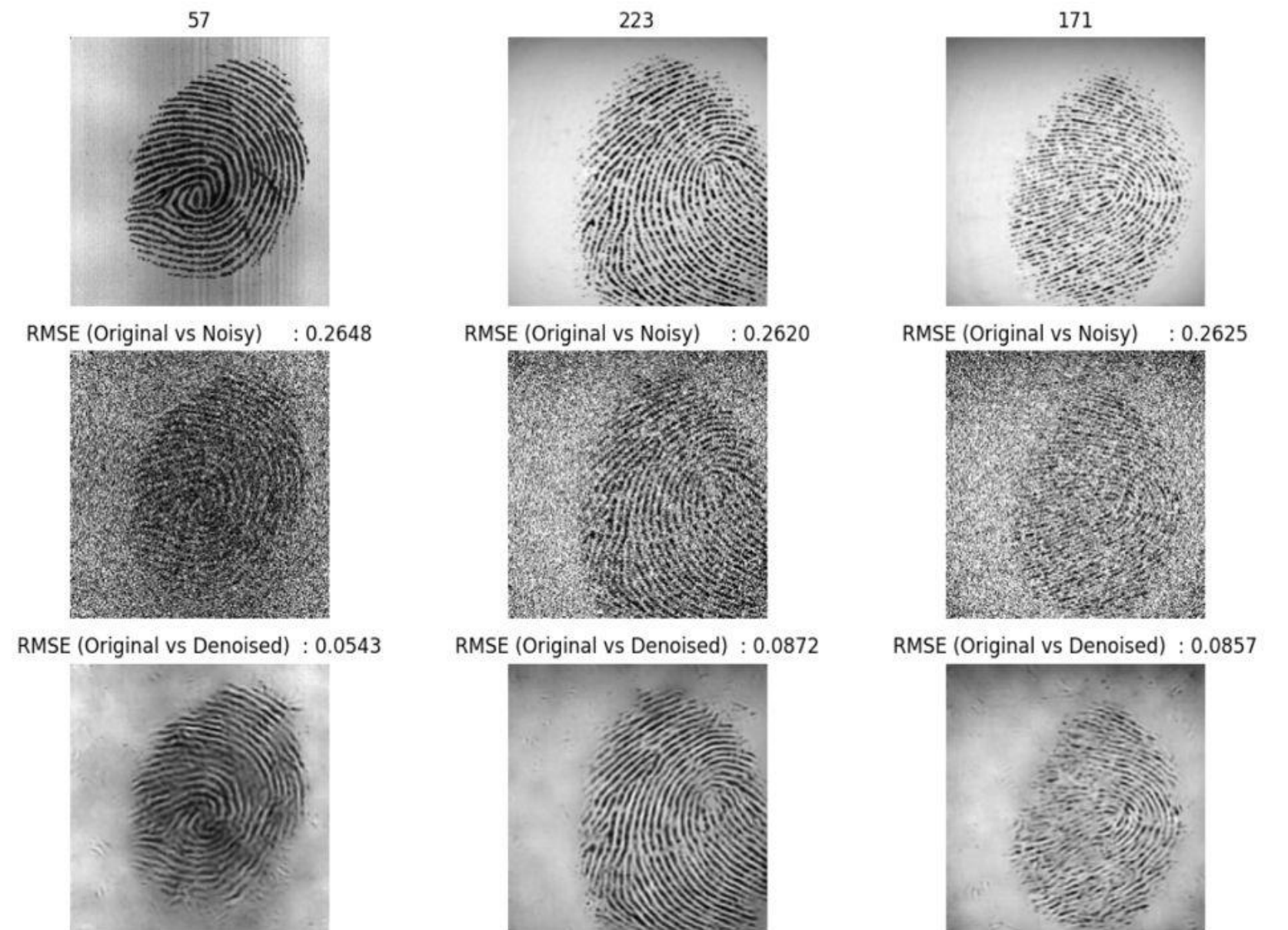
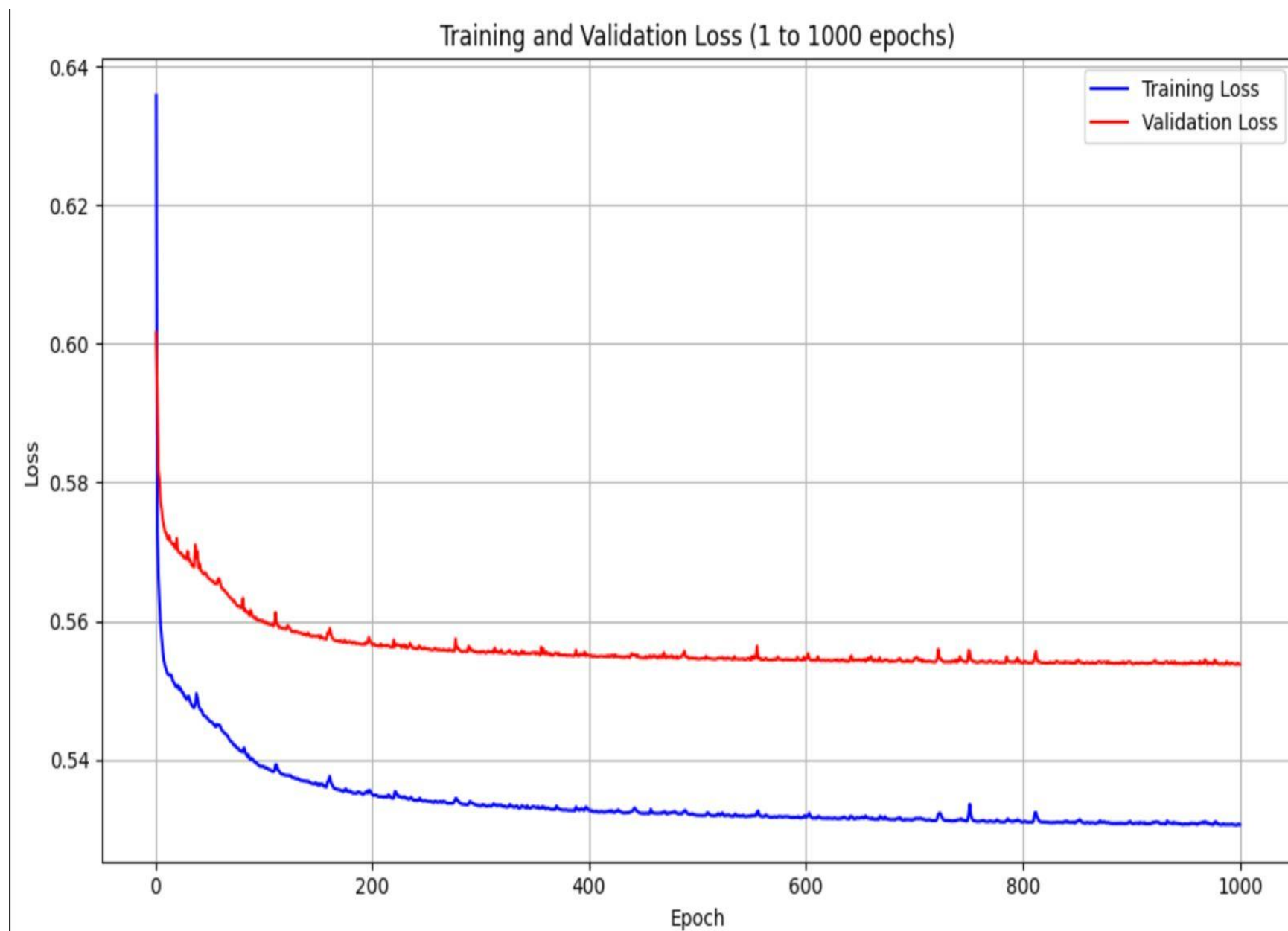
Challenges in using Deep Convolutional Autoencoder:

- ❑ **Model Limitations:** The deep convolutional autoencoders **failed to preserve fine ridge** and minutiae details due to low resolution and suboptimal architecture.
- ❑ **Overfitting Issues:** Lack of regularization and data augmentation led to **overfitting and poor generalization**, evident from the validation loss spike around epoch 80.

Deep Convolutional Autoencoder V1



Deep Convolutional Autoencoder V2



References

- **Reconstruct fingerprint images using deep learning and sparse autoencoder algorithms**
Sergio Saponara, Abdussalam Elhanashi, Alessio Gagliardi
- **Recreating Fingerprint Images by Convolutional Neural Network Autoencoder Architecture.****Sergio Saponara; Abdussalam Elhanashi; Qinghe Zheng**
- **Anomaly Detection With Convolutional Autoencoders for Fingerprint Presentation Attack Detection.** **Jascha Kolberg; Marcel Grimmer; Marta Gomez-Barrero; Christoph Busch**
- **Learning Fingerprint Reconstruction: From Minutiae to Image.****Kai Cao; Anil K. Jain**

THANK YOU

Here is the github link to our project :

<https://github.com/Vishesh-Goyal7/fingerprintAutoencoders>